Characterization of Reachable States and Complete Solver for the Level 2 Menger Twist Cube

Menger Cube Project

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Abstract

For the Level 2 Menger Twist Cube (400 cells), we completely determine the move groups generated by legal atomic moves and prove the completeness of a solver that restores every reachable state to the solved state. We treat two move systems: the standard system (468 atomic moves, generating group G) and the expanded system obtained by adjoining single-layer square-ring rotations (522 atomic moves, generating group H). Our main results are threefold: (1) A state is reachable if and only if its orbitwise permutation parity vector lies in a specified subspace and the twist sums on the two corner orbits together with the flip parity on one edge orbit all vanish. (2) $|G| \approx 2.66 \times 10^{575}$ and $[H : G] = 2$. (3) In both move systems, the implementation `level2-certified` constructs a valid solution consisting solely of legal atomic moves for every reachable state. The proof separates into finitely many deterministic, exhaustive computer checks (using no randomness) and mathematical deductions from those checks.

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1 Introduction

1.1 Problem and Main Results

The Level 2 puzzle consists of the 400 unit cubes (cells) belonging to the Menger sponge within a $9 \times 9 \times 9$ grid. A state obtained by applying a finite sequence of legal rotations (atomic moves) from the solved state is called *reachable*. The goal of this paper is to characterize the set of reachable states via explicit invariants and to provide a complete procedure that solves every reachable state.

Prior work on Level 2 only established constructions in an idealized rotation model prior to collision detection; completeness under the physical collision model (permitting only rotations whose swept volume does not intersect stationary cells) remained unproven. While the existing solver `level2-slice-reduction` solves typical scrambled instances, it relies on heuristic selection rules that preclude a direct proof of completeness on their own. Here we construct a certified solver specifically designed for mathematical verification, prove its completeness, and deduce the completeness of the hybrid solvers as a corollary.

1.2 Proof Architecture

The proof establishes a two-way inclusion between the set G_S of reachable states and the set U_S of states satisfying the invariants:

- $G_S \subseteq U_S$: We verify across all atomic moves that every generator preserves the invariants (Section 4).
- $U_S \subseteq G_S$: For any state in U_S, we explicitly construct a sequence of atomic moves that returns it to the solved state (Section 6).

The construction proving $U_S \subseteq G_S$ is itself the solving algorithm, so proving its termination and correctness directly proves solver completeness.

The computational facts relied upon by the proof are cataloged in Section 8 as Checks C1–C9. Every check is an exhaustive enumeration over a finite set with no probabilistic components.

2 Model and Definitions

2.1 Positions, Orientations, and States

Let $P \subseteq \{-4, \ldots, 4\}^3$ denote the set of Level 2 cell positions, consisting of all points $p = 3b + o$ with $b, o \in \{-1, 0, 1\}^3$ such that each of b and o has at most one zero coordinate. Thus $|P| = 20^2 = 400$.

Let C denote the rotation group of the cube (the group of 3×3 orthogonal matrices with entries in $\{0, \pm1\}$ and determinant 1), of order $|C| = 24$.

Each cell is identified by its position in the solved state (its *home* position). A *state* is a pair $s = (\sigma, \rho)$, where $\sigma\colon P \to P$ is a bijection giving the current position $\sigma(x)$ of the cell whose home is x, and $\rho\colon P \to C$ assigns the current orientation $\rho(x)$ of that cell relative to the identity matrix I in the solved state. The solved state is $e = (\mathrm{id}, I)$.

2.2 Atomic Moves

An *atomic move* a is specified by a rotating region $D_a \subseteq P$, a rotation axis parallel to one of the coordinate axes, and an angle $\theta \in \{90°, -90°, 180°\}$. It induces a position permutation $\pi_a\colon P \to P$ mapping D_a to itself (and acting as the identity outside D_a) together with a rotation matrix $R_a \in C$. Applying a to a state $s = (\sigma, \rho)$ yields $s \cdot a = (\sigma', \rho')$ defined by

$$\sigma'(x) = \pi_a(\sigma(x)), \qquad \rho'(x) = \begin{cases} R_a\,\rho(x) & (\sigma(x) \in D_a), \\ \rho(x) & (\sigma(x) \notin D_a). \end{cases}$$

A sequence of atomic moves (a *word*) $a_1 a_2 \cdots a_k$ is applied from left to right.

Whether an atomic move is legal depends solely on which grid positions are occupied by cells. Because every legal atomic move preserves the occupied set P, legality is independent of the puzzle state. Consequently, the set

S of legal atomic moves is a fixed finite set closed under inversion (reversing the sign of the rotation angle on the same region).

Definition 2.1 (Move Systems)**.**

- *Standard system* S_{std}: all legal frame rotations, recursive expanded rotations, and thin-slab rotations. Here $|S_{\text{std}}| = 468$ (156 regions $\times$ 3 angles).
- *Expanded system* S_{in}: S_{std} adjoined with the 18 inner and outer single-layer square-ring regions on planes $-3, 0, +3$ along each axis. Here $|S_{\text{in}}| = 522$.

The 18 additional regions in S_{in} were obtained by exhaustively enumerating all single-layer cell subsets rotatable around the center of their bounding square and comprise all legal single-layer square regions outside S_{std} (`docs/research/square-region-rotation-model.md`). Collision freedom (zero positive-volume intersection with stationary cells throughout the continuous rotation) for all moves in both systems is certified via exact integer arithmetic in that document.

2.3 Group Formulation

Extending the right action $s \cdot a$ of atomic moves to arbitrary words gives a group action on the set of all states. For a move system S, let G_S denote the set of states reachable from the solved state e by words over S. Since the state transformation induced by a word depends only on the resulting state $s = e \cdot w$, G_S carries a natural group structure in bijection with reachable states. Throughout, we write $G = G_{S_{\text{std}}}$ and $H = G_{S_{\text{in}}}$. Since $S_{\text{std}} \subseteq S_{\text{in}}$, we have $G \leq H$.

3 Position Orbits and Orientation Freedom

3.1 Position Orbits

Define a graph with vertex set P and edges connecting x and y whenever $\pi_a(x) = y$ for some $a \in S$. Its connected components are the *position orbits*, which coincide with the orbits of G_S acting on P.

Proposition 3.1 (Check C2)**.** *In both the standard and expanded systems, there are 11 position orbits, listed in Table 1.*

The class column classifies $p = 3b + o$ by its digit pattern (C has no zero coordinates; E has one zero coordinate; EEa has zero coordinates in b and o along the same axis; EEo along different axes).

Table 1: Position orbits at Level 2. "Orientation group" is the stabilizer H_0 from Section 3.2; "Rolls" is the number of cells admitting single-cell atomic rotations.

Orbit	Size	Representative	Class	Orientation group	Rolls
Ω_0	8	$(-4,-4,-4)$	CC	C_3	0
Ω_1	24	$(-4,-4,-3)$	CE	C_4	24
Ω_2	24	$(-4,-4,-2)$	CC	1	0
Ω_3	96	$(-4,-4,-1)$	EC	1	0
Ω_4	48	$(-4,-4,0)$	EEa	D_4	48
Ω_5	24	$(-4,-3,-2)$	CE	1	0
Ω_6	96	$(-4,-3,-1)$	EEo	1	0
Ω_7	24	$(-4,-3,2)$	CE	1	0
Ω_8	24	$(-4,-2,-2)$	CC	1	0
Ω_9	24	$(-3,-2,-2)$	CE	1	0
Ω_{10}	8	$(-2,-2,-2)$	CC	C_3	0

3.2 Orientation Freedom

Fix a representative position x_Ω for each orbit Ω. Tracking a single cell with home x_Ω, its (position, orientation) pair moves within the finite set $\Omega \times C$. Let $Q_\Omega \subseteq \Omega \times C$ be the set of pairs reachable from (x_Ω, I), and write its fiber over $p \in \Omega$ as $Q_p = \{R : (p, R) \in Q_\Omega\}$. Then $H_0 = H_0(\Omega) = Q_{x_\Omega}$ is a subgroup of C, as the composition of transformations returning the cell to its home position also returns it to its home position.

For each position $p \in \Omega$, choose a *reference orientation* $r_p \in Q_p$ with $r_{x_\Omega} = I$.

Lemma 3.2 (Check C3)**.** *For every orbit Ω and every $p \in \Omega$, $Q_p = r_p H_0$. The isomorphism types of H_0 are as listed in Table 1.*

While $Q_p = r_p H_0$ also follows from general group-action principles (Q_p is invariant under right multiplication by H_0, and comparing paths $(x_\Omega, I) \to (p, r_p)$ and $(x_\Omega, I) \to (p, R)$ yields $r_p^{-1} R \in H_0$), we verify the fibers and subgroup types directly by exhaustive computation.

Definition 3.3 (Twist Coordinate)**.** For a state $s = (\sigma, \rho)$ and a cell with home $x \in \Omega$, define its *twist coordinate* by

$$\tau_s(x) = r_{\sigma(x)}^{-1} \, \rho(x) \, r_x \in C.$$

In the solved state, $\tau_e(x) = I$.

Definition 3.4 (Admissible State)**.** A state s is *admissible* if for every $x \in P$, $\sigma(x)$ lies in the same orbit Ω as x and $\tau_s(x) \in H_0(\Omega)$. Let W denote the set of all admissible states.

Because an element of W is specified by an independent permutation on each orbit Ω_i and an independent choice of $\tau \in H_0(\Omega_i)$ for each cell, we have

$$|W| = \prod_{i=0}^{10} |\Omega_i|! \cdot |H_0(\Omega_i)|^{|\Omega_i|} = (8!)^2 (24!)^6 \, 48! \, (96!)^2 \cdot 3^{16} \cdot 4^{24} \cdot 8^{48}. \quad (1)$$

Lemma 3.5 (Action of Atomic Moves on Twist Coordinates; Check C4, Part 1)**.** *For an atomic move a and a position $p \in \Omega$, define*

$$\delta(a, p) = r_{\pi_a(p)}^{-1} \, R_a(p) \, r_p,$$

where $R_a(p) = R_a$ if $p \in D_a$ and I otherwise. Then $\delta(a, p) \in H_0(\Omega)$, and for the cell x occupying position $p = \sigma(x)$ in state s,

$$\tau_{s \cdot a}(x) = \delta(a, p) \, \tau_s(x).$$

In particular, $W \cdot a \subseteq W$, and hence $G_S \subseteq W$.

For a word $g = a_1 \cdots a_k$, writing $\pi_g = \pi_{a_k} \circ \cdots \circ \pi_{a_1}$ and

$$\delta(g, p) = \delta(a_k, p_{k-1}) \cdots \delta(a_2, p_1) \, \delta(a_1, p_0), \qquad p_0 = p, \; p_j = \pi_{a_j}(p_{j-1}),$$

we have $\delta(g, p) \in H_0(\Omega)$, $\tau_{s \cdot g}(x) = \delta(g, p) \, \tau_s(x)$, and $\delta(g^{-1}, \pi_g(p)) = \delta(g, p)^{-1}$.

Proof. With $\sigma(x) = p$, we compute

$$\tau_{s \cdot a}(x) = r_{\pi_a(p)}^{-1} R_a(p) \rho(x) r_x = r_{\pi_a(p)}^{-1} R_a(p) r_p \cdot r_p^{-1} \rho(x) r_x = \delta(a, p) \tau_s(x).$$

Membership $\delta(a, p) \in H_0(\Omega)$ is verified computationally across all atomic moves and positions. From $e \in W$ and $W \cdot a \subseteq W$ it follows that $G_S \subseteq W$. The claims for words follow by induction on k, and the inversion formula follows from $\delta(g^{-1}, \pi_g(p)) \, \delta(g, p) = I$ since gg^{-1} acts trivially. □

4 Invariants

4.1 Orbit Parity

For an admissible state s, σ preserves each orbit Ω_i. Let $\varepsilon(s) \in \mathbf{F}_2^{11}$ be the vector whose i-th component is the sign parity of $\sigma|_{\Omega_i}$ (0 for even, 1 for odd). By multiplicativity of permutation signs, $\varepsilon(s \cdot a) = \varepsilon(s) + \varepsilon(a)$, where $\varepsilon(a)$ is the parity vector of π_a.

Definition 4.1. For a move system S, define $V_S = \operatorname{span}\{\varepsilon(a) : a \in S\} \subseteq \mathbf{F}_2^{11}$.

Proposition 4.2 (Check C5)**.** *We have* $\dim V_{S_{\mathrm{std}}} = 5$ *and* $\dim V_{S_{\mathrm{in}}} = 6$*. Writing* u_i *for the* i*-th standard basis vector corresponding to* Ω_i*, bases are given by*

$$\begin{aligned} V_{S_{\mathrm{std}}} &= \mathrm{span}\{\, u_0 + u_5 + u_7 + u_8,\ u_1 + u_5 + u_7 + u_9, \\ &\qquad u_2 + u_5 + u_7 + u_{10},\ u_4,\ u_3 + u_6 \,\}, \\ V_{S_{\mathrm{in}}} &= V_{S_{\mathrm{std}}} \oplus \mathrm{span}\{u_9\}. \end{aligned}$$

4.2 Twist Sums and Flip Parity

On the two orbits Ω_0 and Ω_{10} with orientation group C_3, fix an isomorphism $\chi\colon H_0 \to \mathbf{Z}/3$. On the orbit Ω_4 with orientation group D_4, let $C_4 \le H_0$ be the cyclic subgroup of order 4, and let $\chi\colon H_0 \to H_0/C_4 \cong \mathbf{Z}/2$ be the natural quotient homomorphism. A cell with $\chi(\tau) = 1$ is said to be *flipped*.

Definition 4.3. For an admissible state s, define

$$\theta_0(s) = \sum_{x \in \Omega_0} \chi(\tau_s(x)) \in \mathbf{Z}/3, \qquad \theta_{10}(s) = \sum_{x \in \Omega_{10}} \chi(\tau_s(x)) \in \mathbf{Z}/3,$$
$$\varphi(s) = \sum_{x \in \Omega_4} \chi(\tau_s(x)) \in \mathbf{Z}/2.$$

We call θ_0, θ_{10} the *twist sums* and φ the *flip parity*.

Lemma 4.4 (Invariance; Check C4, Part 2)**.** *For every atomic move* a *in either move system and every* $\Omega \in \{\Omega_0, \Omega_4, \Omega_{10}\}$*,*

$$\sum_{p \in \Omega} \chi(\delta(a, p)) = 0.$$

Consequently, for every admissible state s*,* $\theta_0(s{\cdot}a) = \theta_0(s)$*,* $\theta_{10}(s{\cdot}a) = \theta_{10}(s)$*, and* $\varphi(s \cdot a) = \varphi(s)$*.*

Proof. The first equation is verified computationally across all atomic moves ($468 \times 3 = 1404$ pairs in the standard system and $522 \times 3 = 1566$ pairs in the expanded system). The second claim follows from Lemma 3.5 and the fact that χ is a homomorphism:

$$\sum_{x \in \Omega} \chi(\tau_{s \cdot a}(x)) = \sum_{x \in \Omega} \chi(\tau_s(x)) + \sum_{p \in \Omega} \chi(\delta(a, p)),$$

since $x \mapsto \sigma(x)$ is a bijection on Ω. □

4.3 Necessary Conditions

Definition 4.5. For a move system S, define

$$U_S = \{\, s \in W :\ \varepsilon(s) \in V_S,\ \theta_0(s) = 0,\ \theta_{10}(s) = 0,\ \varphi(s) = 0 \,\}.$$

Proposition 4.6. $G_S \subseteq U_S$.

Proof. Clearly $e \in U_S$. If $s \in U_S$, then $s \cdot a \in W$ by Lemma 3.5, $\varepsilon(s \cdot a) = \varepsilon(s) + \varepsilon(a) \in V_S$, and the twist sums and flip parity are preserved by Lemma 4.4. Induction on word length yields $G_S \subseteq U_S$. □

5 Macro Tools

Throughout this section, X_k denotes the 90° rotation around the X-axis of the 1-cell-thick frame at layer $x = k$, and Y_k similarly for the Y-axis. All such moves belong to S_{std}. The commutator of words A, B is written $[A, B] = ABA^{-1}B^{-1}$ (applied left to right).

Definition 5.1. A word T is *pure* if the transformation it induces fixes both the positions and orientations of all cells outside a finite set of positions. The minimal such set of positions is denoted $\operatorname{supp} T$.

Lemma 5.2 (Conjugation)**.** *Let T be a pure word and g an arbitrary word with position action π_g. Then $g^{-1}Tg$ is pure with $\operatorname{supp}(g^{-1}Tg) = \pi_g(\operatorname{supp} T)$. If T maps position $x \mapsto y$, then $g^{-1}Tg$ maps $\pi_g(x) \mapsto \pi_g(y)$.*

Proof. A cell at position $q \notin \pi_g(\operatorname{supp} T)$ is carried outside $\operatorname{supp} T$ by g^{-1}, is fixed in both position and orientation by T, and is returned to q by g with its orientation changes from g^{-1} and g canceling. The position mapping on $\pi_g(\operatorname{supp} T)$ follows immediately from composing the maps. □

5.1 Pure 3-Cycles

Proposition 5.3 (Check C6)**.** *For each orbit Ω_i, the 8-move word in Table 2,*

$$T_i = [\, a_i,\ b_i c_i b_i^{-1} \,],$$

is pure, its support $\operatorname{supp} T_i$ consists of exactly three points of Ω_i, and T_i acts as a 3-cycle on those three points.

Table 2: Pure 3-cycles $T_i = [a_i,\, b_i c_i b_i^{-1}]$ for each orbit.

Orbit	a_i	b_i	c_i	Orbit	a_i	b_i	c_i
Ω_0	Y_{+4}	X_{-4}	Y_{-4}	Ω_6	Y_{-1}	X_{-4}	Y_{-3}
Ω_1	Y_{-3}	X_{-4}	Y_{-4}	Ω_7	Y_{+2}	X_{-4}	Y_{-3}
Ω_2	Y_{-2}	X_{-4}	Y_{-4}	Ω_8	Y_{+2}	X_{-4}	Y_{-2}
Ω_3	Y_{-1}	X_{-4}	Y_{-4}	Ω_9	Y_{+2}	X_{-3}	Y_{-2}
Ω_4	Y_0	X_{-4}	Y_{-4}	Ω_{10}	Y_{+2}	X_{-2}	Y_{-2}
Ω_5	Y_{-2}	X_{-4}	Y_{-3}				

This commutator has the same form as the classic Rubik's Cube corner 3-cycle $[R,\, U^{-1}L^{-1}U]$, exploiting the fact that layer a_i and the b_i-image of layer c_i intersect in a single cell of Ω_i.

Proposition 5.4 (3-Transitivity; Check C7)**.** *For each orbit Ω, $G_{S_{\mathrm{std}}}$ acts transitively on the set of ordered triples of distinct points of Ω (of size $|\Omega|(|\Omega|-1)(|\Omega|-2)$).*

Proof. We perform an exhaustive breadth-first search on the finite graph whose vertices are ordered triples of distinct points in Ω and whose edges are induced by atomic moves, starting from the support triple of T_i. For $|\Omega| = 8, 24, 48, 96$, the search visits all 336, 12,144, 103,776, and 857,280 ordered triples, respectively. □

Corollary 5.5. *For any orbit Ω and any three distinct points $x, y, z \in \Omega$, there exists a word in S_{std} that cycles $x \mapsto y \mapsto z \mapsto x$ and fixes the position and orientation of every other cell.*

Proof. Let T_i cycle $x_0 \mapsto y_0 \mapsto z_0 \mapsto x_0$. By Proposition 5.4, there is a word g with $\pi_g(x_0, y_0, z_0) = (x, y, z)$, and by Lemma 5.2 the conjugate $g^{-1}T_i g$ has the required action. □

5.2 Orientation Tools

Proposition 5.6 (Checks C6, C7)**.** *Each of the following 20-move words fixes all cell positions and is a pure word supported on exactly two points of a single orbit:*

$$\begin{aligned} P_0 &= (Y_{-4}\, T_0\, Y_{-4}^{-1})\,(X_{-4}\, T_0^{-1}\, X_{-4}^{-1}) && (\Omega_0),\\ P_{10} &= (Y_{-2}\, T_{10}\, Y_{-2}^{-1})\,(X_{-2}\, T_{10}^{-1}\, X_{-2}^{-1}) && (\Omega_{10}),\\ P_4 &= (Y_{-4}^{2}\, T_4\, Y_{-4}^{2})\,(X_0\, T_4^{-1}\, X_0^{-1}) && (\Omega_4). \end{aligned}$$

On their two support cells, P_0 and P_{10} left-multiply the twist coordinates by elements with nonzero χ-values summing to $0 \in \mathbf{Z}/3$. The word P_4 left-multiplies the twist coordinates of its two support cells by elements both having $\chi = 1$ (flips). Furthermore, $G_{S_{\mathrm{std}}}$ acts transitively on ordered pairs of distinct points in each of Ω_0, Ω_4, and Ω_{10}.

Lemma 5.7 (Conjugation of Orientation Tools)**.** *Let $\Omega \in \{\Omega_0, \Omega_4, \Omega_{10}\}$, let P be the corresponding tool from Proposition 5.6 with $\operatorname{supp} P = \{x_0, y_0\}$, and let $x \neq y$ be any two distinct points of Ω. There exists a word g with $\pi_g(x_0) = x$ and $\pi_g(y_0) = y$ such that $Q = g^{-1}Pg$ fixes all positions and has $\operatorname{supp} Q = \{x, y\}$. Moreover, if $d_0 \in H_0$ is the element by which P left-multiplies the twist coordinate at x_0, then the element d by which Q left-multiplies the twist coordinate at x is conjugate to d_0 in H_0 and satisfies $\chi(d) = \chi(d_0) \neq 0$.*

Proof. The existence of g follows from 2-transitivity in Proposition 5.6, and $\operatorname{supp} Q = \{x, y\}$ with all positions fixed follows from Lemma 5.2. Tracking the cell at x: g^{-1} carries it to x_0 and left-multiplies its twist coordinate by $\delta(g^{-1}, x) = \delta(g, x_0)^{-1}$; P fixes its position and left-multiplies by d_0; g returns it to x and left-multiplies by $\delta(g, x_0)$. Thus

$$d = \delta(g, x_0)\, d_0\, \delta(g, x_0)^{-1},$$

with $\delta(g, x_0) \in H_0$ by Lemma 3.5. Since χ maps into an abelian group, $\chi(d) = \chi(d_0) \neq 0$ by Proposition 5.6. □

Proposition 5.8 (Single-Cell Rolls; Check C8)**.** *For every position p in Ω_1 and Ω_4, there are three atomic moves ($\pm 90°, 180°$) that rotate only the single cell at p. Together with the identity, the elements by which they left-multiply the twist coordinate at p form the full group H_0 on Ω_1 and the subgroup $C_4 \leq H_0$ on Ω_4.*

6 Main Theorem and Solving Algorithm

Theorem 6.1 (Characterization of Reachable States)**.** *For $S \in \{S_{\text{std}}, S_{\text{in}}\}$, we have $G_S = U_S$. That is, a state s is reachable under S if and only if all four of the following conditions hold:*

1. *s is admissible (every cell lies in its home orbit and has twist coordinate in H_0).*
2. *$\varepsilon(s) \in V_S$.*
3. *$\theta_0(s) = \theta_{10}(s) = 0$.*
4. *$\varphi(s) = 0$.*

Necessity was established in Proposition 4.6. We prove sufficiency by constructing a word that reduces any $s \in U_S$ to the solved state e.

6.1 Algorithm L2-C

Given an input state s:

Phase 0 (Certification) Check Conditions 1–4. If any condition fails, report that s is unreachable and terminate.

Phase 1 (Parity Normalization) Find a word w over S with $\varepsilon(w) = \varepsilon(s)$ via breadth-first search on the Cayley graph of V_S generated by $\{\varepsilon(a) : a \in S\}$, and apply w.

Phase 2 (Position Restoration) For each orbit Ω, repeat until all cells of Ω are at home: Pick a position a occupied by a cell not at home, and let $b \neq a$ be that cell's home. Let c be the home of the cell currently at b; if $c = a$, instead choose $c \notin \{a, b\}$ to be any other position in Ω whose occupant is not at home. Apply the pure 3-cycle from Corollary 5.5 cycling $a \mapsto b \mapsto c \mapsto a$.

Phase 3 (Orientation Restoration) All cells are now at their home positions.

(a) For each $\Omega \in \{\Omega_0, \Omega_{10}, \Omega_4\}$, order its positions as $x_1, \ldots, x_n$. For $i = 1, \ldots, n-1$ in order, while $\chi(\tau(x_i)) \neq 0$, apply the conjugate pair tool from Lemma 5.7 supported on $\{x_i, x_{i+1}\}$.

(b) At each position of Ω_1 and Ω_4 where $\tau \neq I$, apply the single-cell roll from Proposition 5.8 that sets $\tau = I$.

6.2 Proof of Correctness

Proof of Sufficiency for Theorem 6.1. Let $s \in U_S$.

Phase 1. Since $\varepsilon(s) \in V_S$ and V_S is spanned by $\{\varepsilon(a) : a \in S\}$, a word w with $\varepsilon(w) = \varepsilon(s)$ exists. The state $s_1 = s \cdot w$ satisfies $\varepsilon(s_1) = 0$, and by Lemmas 3.5 and 4.4 we still have $s_1 \in W$ and $\theta_0 = \theta_{10} = \varphi = 0$.

Phase 2. On each orbit Ω, $\sigma|_\Omega$ is an even permutation. The cell at a has home $b \neq a$, and the cell at b cannot have home b (since the cell with home b is at a). If the cell at b has home $c \neq a$, then the cell with home c is at b, so the occupant of c is not at home (and $c \neq b$). If $c = a$, the cells at a and b occupy each other's homes; because $\sigma|_\Omega$ is even, it cannot be a single transposition, so another misplaced cell exists at some $c \notin \{a, b\}$. In either case, none of the cells at a, b, c are at home, and applying the 3-cycle $a \mapsto b \mapsto c \mapsto a$ places the cell from a into its home b. Because the word from Corollary 5.5 fixes all cells outside $\{a, b, c\}$, the number of solved positions in Ω increases by at least 1 while all other orbits remain untouched. Thus Ω is solved in at most $|\Omega|$ steps. Each 3-cycle is even, preserving $\varepsilon = 0$, and lies in G, preserving W and $\theta_0 = \theta_{10} = \varphi = 0$. At the end of Phase 2, the resulting state s_2 satisfies $\sigma = \mathrm{id}$, $s_2 \in W$, and $\theta_0 = \theta_{10} = \varphi = 0$.

Phase 3. On the seven orbits with trivial orientation group $H_0 = 1$, $\tau = I$ and hence $\rho = I$ already.

For Ω_0 and Ω_{10}: the tool from Lemma 5.7 multiplies $\tau(x_i)$ by an element d with $\chi(d) \neq 0$. Since $H_0 \cong \mathbf{Z}/3$, at most two applications set $\tau(x_i) = I$. Because the tool is supported on $\{x_i, x_{i+1}\}$, the previously solved orientations at $x_1, \ldots, x_{i-1}$ remain I. Upon completing $i = n - 1$, the invariant $\theta = 0$ forces $\chi(\tau(x_n)) = 0$, so $\tau(x_n) = I$.

For Ω_4: the tool from Lemma 5.7 multiplies $\tau(x_i)$ by an element with $\chi(d) = 1$, so a single application clears $\chi(\tau(x_i)) = 0$. Upon completing

$i = n - 1$, the invariant $\varphi = 0$ forces $\chi(\tau(x_n)) = 0$. At this point every cell in Ω_4 has $\tau \in C_4$, and the single-cell rolls of Proposition 5.8 independently reduce each cell to $\tau = I$.

For Ω_1: here $H_0 \cong C_4$, so the single-cell rolls of Proposition 5.8 independently reduce each cell to $\tau = I$.

When all positions are solved, $\tau(x) = r_x^{-1}\rho(x)r_x$, so $\tau = I$ is equivalent to $\rho = I$. Thus we reach the solved state e. Concatenating all applied words into v gives $s \cdot v = e$, so $s = e \cdot v^{-1} \in G_S$. □

Remark 6.2 (Expanded System). Every tool used in Phases 2 and 3 belongs to S_{std}. The only difference between S_{std} and S_{in} lies in the subspace V_S checked in Phase 0 and the parity word w applied in Phase 1. In the expanded system, w contains at most one square-ring rotation.

6.3 Corollaries

Corollary 6.3 (Group Order)**.**

$$|G_S| = \frac{|W| \cdot |V_S|}{2^{11} \cdot 3 \cdot 3 \cdot 2},$$

where $|W|$ *is given by* (1)*. Numerically,*

$$|G| = \frac{|W| \cdot 2^5}{36864} \approx 2.6557 \times 10^{575} \quad (576 \textit{ digits}), \qquad |H| = 2\,|G|.$$

Proof. Consider the map $\Phi = (\varepsilon, \theta_0, \theta_{10}, \varphi)\colon W \to M = \mathbf{F}_2^{11} \times \mathbf{Z}/3 \times \mathbf{Z}/3 \times \mathbf{Z}/2$. An element of W is determined by independent orbit permutations and cell twists $\tau \in H_0$. Since ε depends only on the permutations, the remaining components depend only on τ, and each homomorphism χ is surjective, every fiber of Φ has size $|W|/|M|$. Because $U_S = \Phi^{-1}(V_S \times \{0\})$, we obtain $|U_S| = |W| \cdot |V_S|/|M|$ with $|M| = 2^{11} \cdot 18 = 36{,}864$. □

Corollary 6.4 (Expanded vs. Standard System)**.** $[H : G] = 2$*. A state* s *reachable in* H *is reachable in* G *if and only if* $\varepsilon(s) \in V_{S_{\text{std}}}$*.*

This agrees with the independent proof via Schreier generator rewriting in `docs/research/square-region-rotation-impact.md`.

Corollary 6.5 (Membership Test)**.** *Reachability of a given state can be decided in time linear in the number of cells by checking Conditions 1–4.*

Corollary 6.6 (Pure Orientation Subgroup)**.** *The subgroup of reachable states with all cells at their home positions is the subgroup of* $\prod_x H_0$ *cut out by* $\theta_0 = \theta_{10} = \varphi = 0$*, of order* $3^{14} \cdot 4^{24} \cdot 8^{48}/2$*, and is identical in both move systems.*

7 Completeness of the Implementation

Theorem 7.1 (Completeness of `level2-certified`)**.** *In both the standard and expanded move systems, the implementation* `level2-certified` *(*`packages/solver-core/src/algorithms/level2CertifiedSolver.ts`*) satisfies:*

1. *If the input state is reachable, it outputs a finite word of legal atomic moves that transforms the input into the solved state.*
2. *If the input state is unreachable, it outputs no word and reports which invariant condition is violated.*

(Assuming the host machine does not suffer hardware or out-of-memory faults.)

Proof. The implementation executes Algorithm L2-C directly. The conjugating words g required in Phases 2 and 3 are obtained by breadth-first search on the graph of ordered triples (or ordered pairs) starting from the base tool's support. This search imposes no depth or vertex cap and verifies at runtime that every vertex is reached (throwing an error if the visited count differs from $n(n-1)(n-2)$ or $n(n-1)$). When the tool library is initialized, each base tool word is re-verified for purity and support. Thus every step in the proof of Theorem 6.1 holds verbatim inside the implementation.

Before returning a solution, the implementation replays the word move-by-move on a copy of the caller's state, verifying legality prior to each move and checking at the end that all 400 cells match the solved positions and orientations (soundness). Completeness follows from Theorem 6.1, which guarantees that the constructed word always passes replay. □

Adjacent atomic moves on the same region are merged or canceled in post-processing without altering the induced transformation.

Move Counts. Phase 1 uses at most $\dim V_S \leq 6$ moves, Phase 2 uses at most one conjugated 3-cycle (8 moves plus conjugation) per misplaced cell, and Phase 3 uses a bounded number of orientation tools per misoriented cell. Empirically, solutions require roughly 3,700–4,100 moves in both systems, with solve times under 0.1 s after initial table construction. These solutions are not move-optimal.

7.1 Hybrid Short-Solution Solvers

Corollary 7.2. `level2-slice-reduction` *solves every reachable state in the standard system, and* `level2-inner-square-coset` *solves every reachable state in the expanded system.*

Proof. Write $G = G_{S_{\text{std}}}$ and $H = G_{S_{\text{in}}}$. In the standard system, the heuristic short-solution search is finite and replays any candidate word move-by-move with legality checks, accepting it only if it reaches the solved state. If the heuristic search exceeds its budget or fails to reach the solved state, the solver invokes the certified solver of Theorem 7.1 on the original input $s \in G$. In either branch, it returns a finite legal word solving s.

In the expanded system, let $\psi\colon V_{S_{\text{in}}} \to V_{S_{\text{in}}}/V_{S_{\text{std}}} \cong \mathbf{F}_2$ be the quotient map on orbit parity. Because all invariants other than the parity subspace coincide between the two systems and $\dim V_{S_{\text{in}}} - \dim V_{S_{\text{std}}} = 1$, Theorem 6.1 gives $\ker(\psi \circ \varepsilon) = G$ and $[H : G] = 2$. Given $s \in H$, if $\psi(\varepsilon(s)) = 0$ then $s \in G$ and the standard-system solver applies directly. If $\psi(\varepsilon(s)) = 1$, the solver applies a single square-ring atomic move r with $\psi(\varepsilon(r)) = 1$ (which exists because S_{in} spans $V_{S_{\text{in}}}$, and is legal at s because legality is state-independent). Then $\psi(\varepsilon(s \cdot r)) = 0$, so $s \cdot r \in G$ is solved by the standard-system solver. □

7.2 Solving up to a Whole-Cube Rotation

Let K be the group of the 24 proper rotations of a cube. For each coordinate axis, a quarter turn of each of the three coarsest slices rotates every cell once about that axis. These turns are legal standard moves and the three resulting quarter turns generate K. Thus $K \subseteq G_S$ in both move systems, and the goal of the two solvers below is to reach any state in K.

Corollary 7.3. *Both* `level2-rotation-target` *and* `level2-rotation-quotient-search` *solve every reachable state up to a whole-cube rotation, in the standard and expanded move systems.*

Proof. In this subsection, composition is written right to left, so a move a sends s to as. Relabeling the home position of each cell by $R \in K$ and replacing its orientation Q by QR^{-1} represents the state sR^{-1}. Physical moves select cells by current position and act on their orientations from the left, so the same moves apply after relabeling.

The target selection solver scores the 24 choices using positional cycles and orientation defects at cells already in place, and selects one R. Since $sR^{-1} \in G_S$, Theorem 7.1 gives a legal word w with $wsR^{-1} = e$. On the original input, that word ends at $ws = R$.

The search solver uses the right cosets sK as vertices. If the cell at a fixed site has orientation Q, the search key represents sQ^{-1}. For $T \in K$, the state sT has orientation QT at the same site, giving the same key since $sT(QT)^{-1} = sQ^{-1}$. Equal keys also imply membership in the same right coset. Left multiplication by a legal move preserves this equivalence, so the search can keep one representative per key. The solver checks the goal first, then explores one move from each of the input and goal cosets. At a meeting vertex, the inverse move from the goal frontier applies to the input

representative and ends in K. This finds a shortest solution whenever one exists within two moves. Otherwise Theorem 7.1 supplies a solution ending at $e \in K$. Both solvers replay their output on the original input, checking the legality of every move and requiring the endpoint to belong to K. □

8 Summary of Computational Checks

We list the computational checks on which the proof relies. All checks are executed by `research/level2-completeness/certify.ts` using code independent of the solver implementation (sharing only the base tool words). Atomic moves are taken from the engine's legal move enumeration, and all position and orientation calculations use exact integer arithmetic. Run `npm run prove:l2` to regenerate `research/level2-completeness/out/certificate.json`.

Check C1. The sets of legal atomic moves are closed under inversion (468 moves in S_{std}, 522 moves in S_{in}).

Check C2. The position orbits in both systems are the 11 orbits listed in Table 1 (Proposition 3.1).

Check C3. Exhaustive traversal of the (position, orientation) automaton on each orbit confirms that H_0 is a subgroup, $Q_p = r_p H_0$, H_0 has the isomorphism type in Table 1, and χ is a homomorphism (Lemma 3.2).

Check C4. For every atomic move a and position p, $\delta(a, p) \in H_0$, and on $\Omega_0, \Omega_4, \Omega_{10}$, $\sum_p \chi(\delta(a, p)) = 0$ (Lemmas 3.5 and 4.4).

Check C5. The dimension and basis of the parity subspace V_S match Proposition 4.2.

Check C6. Base tool words: $T_0, \ldots, T_{10}$ are pure 3-cycles, and P_0, P_4, P_{10} are position-fixing 2-cell tools with χ-values as stated in Proposition 5.6.

Check C7. Exhaustive verification of 3-transitivity on each orbit and 2-transitivity on $\Omega_0, \Omega_4, \Omega_{10}$ (Propositions 5.4 and 5.6).

Check C8. Every position in Ω_1 and Ω_4 admits three single-cell rolls generating the subgroups stated in Proposition 5.8.

Check C9. Exact computation of the group orders (Corollary 6.3) and the index $|H|/|G| = 2$.

Regression Tests. In addition to the checks above, `npm run test:l2` performs:

- Direct synthesis of states in U_S without applying atomic moves (sampling orbit permutations and cell twists τ and adjusting them to satisfy Conditions 2–4), verifying that every synthesized state is solved. This tests $U_S \subseteq G_S$ on states unlikely to arise from short random scrambles.
- Verification that states violating exactly one of Conditions 1–4 are rejected with the corresponding diagnostic.
- Verification that random scrambles drawn from all legal atomic moves are solved.

These tests guard against implementation regressions; the mathematical guarantee of completeness rests on the proof in Section 6.

9 Development Notes

Relationship to the Heuristic Solver. The solver `level2-slice-reduction` greedily applies macro tools by class, achieving shorter solutions (around 1,300 moves). However, its tool selection and orientation correction rely on empirical rules for which guaranteed progress across all inputs could not be established. We therefore constructed a separate certified algorithm prioritizing provability over move count.

Why Direct Schreier–Sims Enumeration Was Avoided. We considered computing $|G|$ directly via the deterministic Schreier–Sims algorithm. Although a faithful permutation representation requires degree only 840, a base has length near 400 with basic orbits of size up to 384, making deterministic sifting of Schreier generators impractical. Instead, we adopted the classical approach of matching an invariant upper bound $|U_S|$ with a constructive lower bound via pure 3-cycles and orientation tools.

Finding the Invariants. Introducing the twist coordinates τ reveals that each atomic move left-multiplies τ by an element of H_0. Summing the image of τ in the abelianization of H_0 over an orbit yields a quantity that shifts by a constant per atomic move. Evaluating this shift across all atomic moves showed that it vanishes identically on the two C_3 orbits and on the D_4/C_4 quotient. The C_4 components impose no constraint because single-cell rolls exist as legal atomic moves.

Finding the Macro Tools. The pure 3-cycles were found by exhaustively searching commutators $[a, bcb^{-1}]$ over triples (a, b, c) of 1-cell-thick frame moves, yielding 8-move words on all 11 orbits. The orientation tools were obtained as quotients of two conjugates of a single 3-cycle that share the same position action but differ in their orientation action.

Expanded System. Passing from S_{std} to S_{in} increases the dimension of the parity subspace by 1 while leaving all other invariants unchanged, allowing the same proof to cover both systems and independently confirming $[H : G] = 2$.

10 Scope and Limitations

- The theorems in this paper apply to the two move systems defined in Section 2. Thick multi-layer square rotations and rotations around axes not parallel to the coordinate axes belong to neither system, and no claims are made regarding groups adjoined with such moves.
- Collision-free legality of atomic moves is taken as an input from `docs/research/square-region-rotation-model.md`, where both systems are certified via exact integer arithmetic.
- The proof relies on finitely many exhaustive computer checks. While the check script is deterministic and reproducible, it has not been formally verified in a proof assistant; to maximize independence, the verification script and solver implementation are written as separate codebases.
- The constructed solutions are not shortest paths, and no claims are made regarding God's number beyond the information-theoretic lower bound $d \geq \lceil \log_{468} |G| \rceil = 216$ moves.
- For the expanded system at Levels 3 and 4, see `docs/proofs/inner-square-completeness.md`.